

CSC 8228: Privacy Aware Computing - Assignment 2 Full Solutions

1. Local Sensitivity (21 pts)

Theory:

- Local sensitivity measures the maximum change in a function's output by modifying one input.
- For a salary range [10,000, 100,000], the maximum difference is the range itself.

Solution:

1. Identify the range: [10,000, 100,000].
2. Calculate the maximum salary difference: $100,000 - 10,000 = 90,000$.
3. Local sensitivity = Maximum difference = 90,000.

2. Measure Similarity (15 pts)

Theory:

- Nodes in two graphs are paired to maximize similarity. Eccentricity (ecc score) is set to 0.5.

Solution:

1. Start with mapped nodes A -> I, B -> H, C -> G.
2. Calculate similarity scores for all unmapped nodes.
3. Pair nodes iteratively to minimize the eccentricity score, repeating until all nodes are paired.

3. k-Degree Anonymity (15 pts)

Theory:

- The Erdos-Gallai algorithm verifies if a degree sequence can form a simple graph.
- It checks if the sum of degrees satisfies inequalities for all k.

Solution:

1. Input degree sequence in non-increasing order.
2. Apply Erdos-Gallai conditions: $\sum(d_i) \leq k(k-1) + \sum(\min(k, d_j))$ for $k = 1$ to n .
3. Verify the inequality for each k . If true for all k , the sequence is realizable.

4. Earth Mover's Distance (EMD) (13 pts)

Theory:

- EMD calculates the cost of transforming one distribution into another.

Solution:

1. Compare distributions {4K, 4K, 5K, 5K, 6K, 6K} and {7K, 8K, 9K, 10K, 11K, 12K, 13K, 13K}.
2. Match values to minimize cost, e.g., 4K $\rightarrow$ 7K (Cost = $|4K - 7K| = 3K$).
3. Sum all costs to get total EMD.

5. KL-Distance (13 pts)

Theory:

- KL-divergence measures the difference between two probability distributions.
- Formula: $D_{KL}(P \parallel Q) = \sum(P(x) * \log(P(x) / Q(x)))$.

Solution:

1. Compute $P(x) / Q(x)$ for each x .
2. Multiply by $P(x) * \log(P(x) / Q(x))$, then sum results.

6. Exponential Mechanism (13 pts)

Theory:

- Exponential mechanism selects outputs proportional to their utility scores.
- Formula: $P(q_i) = \exp(e * u(x, q) / (2 * \Delta_u))$.

Solution:

1. Utility values: $\{3, 1, 0\}$ for $\{q_1, q_2, q_3\}$. Privacy parameter $\epsilon = 0.5$, $\Delta_u = 1$.
2. Compute $P(q_i)$:
 - $P(q_1) = \exp(0.75)$, $P(q_2) = \exp(0.25)$, $P(q_3) = \exp(0)$.
3. Normalize probabilities by dividing by total $Z = \text{Sum}(\exp(\text{values}))$.

7. Laplace Mechanism (10 pts)

Theory:

- Laplace noise is added to preserve differential privacy.

Solution:

1. True average = 80, privacy parameter $\epsilon = 0.6$.
2. Compute Laplace scale: $b = \Delta_f / \epsilon$ (Δ_f depends on the range of scores divided by the dataset size).
3. Add Laplace noise with scale b to true average: Noisy Average = $80 + \text{Lap}(b)$.